DFT reference vs MLIP prediction | complete energy and force train/test parity

Energy parity

A Scratch-600K

train: n=5,358; ρ=0.89, r=0.90

test: n=801; ρ=0.45, r=0.46

B FT-600K

train: n=5,358; ρ=0.85, r=0.86

test: n=801; ρ=0.47, r=0.44

C Scratch-MultiT

train: n=7,977; ρ=0.89, r=0.93

test: n=2,301; ρ=0.52, r=0.46

D FT-MultiT

train: n=7,977; ρ=0.95, r=0.96

test: n=2,301; ρ=0.43, r=0.44

MLIP $E/N - \langle E/N \rangle_{\text{cluster}}$ (eV atom^{-1})

DFT $E/N - \langle E/N \rangle_{\text{cluster}}$ (eV atom^{-1})

Energy count per hexbin

Force parity

A Scratch-600K

train: n=1,336,032; ρ=0.99, r=0.99

test: n=206,658; ρ=0.84, r=0.80

B FT-600K

train: n=1,336,032; ρ=1.00, r=1.00

test: n=206,658; ρ=0.88, r=0.88

C Scratch-MultiT

train: n=1,983,330; ρ=0.99, r=0.99

test: n=584,658; ρ=0.86, r=0.74

D FT-MultiT

train: n=1,983,330; ρ=1.00, r=1.00

test: n=584,658; ρ=0.90, r=0.90

MLIP force F (eV Å^{-1})

DFT force F (eV Å^{-1})

Force count per hexbin

train test